Multi-agent Interactions

Game theory

Each player maximises their expected payoff.

A player's best action depends on what the others do and what they know.

Payoff matrices

Each player's payoff under each outcome of the game.

	Player 2: C	**Player 2: D**
Player 1: C	$(3, 3)$	$(0, 5)$
Player 1: D	$(5, 0)$	$(1, 1)$

The first number is Player 1's payoff; the second is Player 2's. Here C means cooperate and D means defect.

Nash equilibrium

A Nash equilibrium is a strategy profile in which each player is acting optimally, holding the others' strategies fixed.

In other words

No player can improve their payoff by changing only their own action.

The Prisoner's Dilemma

Defection gives a higher payoff whatever the other player does.

So the unique Nash equilibrium is for both players to defect, even though both would prefer mutual cooperation.

	C	D
C	$(3, 3)$	$(0, 5)$
D	$(5, 0)$	$(1, 1)$

Commitments can enable cooperation

What if the prisoners could commit to their decisions in advance?

If both can commit to cooperating, then they both prefer the resulting $(3, 3)$ outcome to mutual defection at $(1, 1)$.

Chicken

Two cars are driving towards each other, each trying to be the last to swerve.

	Swerve	**Straight**
Swerve	$(2, 2)$	$(1, 3)$
Straight	$(3, 1)$	$(0, 0)$

The two Nash equilibria have one player swerve while the other drives straight.

Commitment races

Suppose a player can commit to driving straight by throwing out their steering wheel.

Each player wants to commit first, forcing the other to swerve. But if both commit, neither can swerve and they crash.

Commitments are a double-edged sword

They can support cooperation, but they can also make conflict unavoidable.

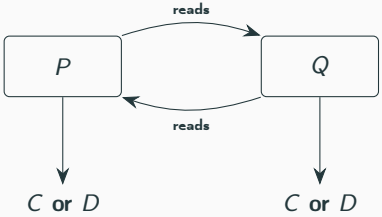
Open-source game theory

Each player submits a program that can inspect the other player's program before choosing an action.

For programs P and Q,

$$P(Q) \in \{C, D\}.$$

A program's source code is therefore a visible, conditional commitment.



FairBot

FairBot cooperates exactly when it can prove that its opponent cooperates with it:

FairBot(Q)

If there is a proof that $Q(\text{FairBot}) = C$, return C.

Otherwise, return D.

With a sound proof system, FairBot cannot be exploited: whenever it cooperates, its opponent cooperates too.

Löb's theorem

Let T extend Peano arithmetic and have computably enumerable axioms. Write $\Box\varphi$ for the sentence asserting that a T-proof of φ exists.

Löb's theorem

For every sentence φ,

$$T \vdash (\Box\varphi \to \varphi) \quad \Longrightarrow \quad T \vdash \varphi.$$

If T proves that provability of φ implies φ, then T already proves φ.

Löb's theorem and FairBot

Use the same theory T and provability operator $\Box$.

Löb's theorem

For every sentence φ,

$$T \vdash (\Box\varphi \to \varphi)$$

implies

$$T \vdash \varphi.$$

FairBot(Q)

If there is a T-proof that

$$Q(\text{FairBot}) = C,$$

return C.

Otherwise, return D.

FairBot here is an idealised proof-existence rule.

Two FairBots cooperate

Let X be the statement

$$\text{FairBot}(\text{FairBot}) = C.$$

By inspecting FairBot's definition, T proves

$$\Box X \to X.$$

Löb's theorem gives $T \vdash X$, so FairBot finds a proof and cooperates.

ε-grounded FairBot

A simulation-based alternative:

εGroundedFairBot(Q)

With probability ε, cooperate.

Otherwise, return Q(εGroundedFairBot).

The small chance of immediate cooperation grounds the recursion. Otherwise, the program responds to what its opponent does against it.

Two desiderata for open-source agents

Non-exploitability

If P cooperates with Q, then Q also cooperates with P:

$$P(Q) = C \implies Q(P) = C.$$

Robust cooperation

Cooperate with copies of yourself, and more generally with agents whose cooperation you can verify.

Leader–follower games

In Chicken, the actions are also called Hawk (straight) and Dove (swerve).

In a leader–follower game, the leader commits first. The follower observes that commitment and chooses a best response.

The leader chooses a commitment while anticipating the follower's response. This can create an advantage over simultaneous choice.

Open-source programs complicate this picture: each commitment can depend on the other, so it may be unclear who leads and who follows.

Interactive game: the program $c(n)$

$c(n)$

Commit at time n if you cannot tell your opponent has already committed.

You can only tell your opponent has already committed if they committed before time $n - 100$.